1. Identity

Document	STAR-PKG-STAR-ORBIT-EN · v1.4 · 2026-10-20
Package	star-orbit
Version	0.2.2
Capability	star.orbit.propagation
Maturity	VERIFIED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_orbit

2. Licence and obligations

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3. What it needs at run time

NEEDS	LICENCE	SOURCE
star-timescales	Apache-2.0	S.T.A.R.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_orbit-
```

```
0.2.2-py3-none-any.whl
```

FILE	SHA-256
star_orbit-0.2.2-py3-none-any.whl	12c1c188db0be4948b53f5efad4380aec8dd0c8d7b0074b3401f1dd1fcaf5913
star_orbit-0.2.2-source.zip	c857a787886c69b746fad8260b226ce975f03a64a5e79e596f4856b2b3f6a90c

5. What it does, and its limits

Cowell numerical orbit propagation (fixed-step RK4) with two-body + J2 gravity, federated with star-timescales for epochs.

Drag/SRP in the released package, higher-order geopotential, third bodies, published reference trajectory comparison, any operational or flight use.

- `propagate_orbit(OrbitState, duration_sec, step_sec, include_j2=True, ...)` → `list[OrbitState]`, SI units (m, s, m/s), mu = 3.986004418e14 m^3/s^2, Re = 6378137 m, J2 = 1.08262668e-3 (WGS-84 / EGM96).
- Requesting a force that is not available (drag/SRP without star-weather) raises `RuntimeError`; a force is never silently dropped.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_orbit_0_2_2,
  author  = {Cavazzini, Andrea},
  title   = {{star-orbit}: Numerical orbit propagation (Cowell, fixed-step RK4) with two-body
+ J2 gravity, linked to star_timescales},
  version = {0.2.2},
```

```
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_orbit/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts